Venkata Lakshmi Parimala Pasupuleti

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AgentProbe: github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA

PROFESSIONAL SUMMARY

ML engineer with a **production fraud analytics background** - applied to fraud/risk roles at Capital One, Intuit, Chime, and Sezzle because fraud detection is where I want to build. Built CrashClarity (full ML lifecycle: feature engineering → model selection → 85%+ accuracy on 100K+ records) and AgentProbe (production ML monitoring, offline experimentation framework, automated regression detection). 8+ years Python in production at Entrata - data pipelines at scale, quality gates, fintech data integrity. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). Demo: *github.com/venkatalakshmiparimala/AgentProbe*

CORE COMPETENCIES

Production ML Monitoring | Offline Experimentation Frameworks | Fraud ML & Risk Analytics | ML Lifecycle (Data → Prod)

Statistical Evaluation | Feature Engineering | Python / FastAPI | SQL (PostgreSQL 8+ yrs) | Large-Scale Data Pipelines

AWS / Docker / CI/CD | LLM Evaluation | Cross-Functional Delivery

EXPERIENCE

Entrata India (Db Xento Systems)

Jan 2017 - Aug 2024

Software Engineer → Associate Software Engineer → Trainee

Pune, India

- Built **production data pipelines and ML infrastructure** on AWS (S3, Glue, EC2) handling **1M+ records** with automated quality checks and data integrity controls - the same discipline as maintaining production fraud ML models in a fintech environment.
- Improved production query performance by **30%** by architecting PostgreSQL schemas across 20+ projects; maintained **99.5% uptime** for 6+ months via TDD/BDD quality gates - zero security incidents, full PHI compliance.
- Eliminated **80% of manual operational workload** by designing and automating decision support workflows (Python + PL/SQL) across 120+ Agile sprints; cross-functional delivery across Product, Engineering, and QA for **100K+ monthly active users**.
- Cut microservices execution time by **50%** via Docker containerization; reduced critical issue backlog by resolving 100+ high-priority production bugs across fintech data management systems.

George Mason University

Aug 2025 - May 2026

Graduate Teaching Assistant - Python & Databases

Fairfax, Virginia

- Delivered Python and database mentoring for 30+ undergraduates; translated statistical modeling concepts across technical and non-technical audiences - the same cross-functional communication needed across Product, Engineering, and Risk.

PROJECTS

CrashClarity: Full ML Lifecycle on 100K+ Records ML LIFECYCLE

Python · R · Random Forest · Logistic Regression · Feature Engineering · Statistical Evaluation · NumPy · Pandas · GitHub Pages

End-to-end production ML pipeline demonstrating the complete lifecycle Signifyd's role requires. **Feature engineering**: extracted meaningful signals from raw data (weather, road conditions, time patterns, urban/rural context) - analogous to extracting fraud signals from transaction and behavioral data. **Model selection**: trained and compared Random Forest vs Logistic Regression with rigorous statistical evaluation. **Production deployment**: interactive dashboard presenting model insights to non-technical stakeholders. **85%+ accuracy** on 100K+ records. *venkatalakshmiparimala.github.io/Portfolio/project_crashclarity.html*

AgentProbe: Production ML Monitoring & Experimentation Platform OPEN SOURCE

Python · LangChain · LangGraph · FastAPI · ChromaDB · React · Anthropic · OpenAI · Groq · Ollama · Docker · CI/CD

Production system built around **offline experimentation and automated model monitoring** - the two core disciplines Signifyd's role requires. **Offline experimentation framework**: runs systematic multi-provider eval experiments with 5 metrics per run; tracks metric trends across runs and detects statistical regressions - equivalent to the offline experimentation frameworks for model development cited in the JD. **Production ML monitoring**: per-run scoring engine flags anomalies and performance regressions before they compound; Auto Fix agent diagnoses root cause (retrieval vs generation vs prompt) and closes the loop. **15-endpoint FastAPI production backend** with Docker, CI/CD, observability dashboard. *github.com/venkatalakshmiparimala/AgentProbe*

AI on FHIR: NLP-Driven Medical Records Query System

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings · NumPy · Pandas

NLP pipeline mapping natural language queries to structured FHIR APIs via LLMs, BERT, and embeddings. HIPAA-compliant RBAC. **Reduced retrieval time ~60%**. *github.com/venkatalakshmiparimala/AI_on_FHIR*

EDUCATION

M.S. Data Analytics Engineering - George Mason University 2024-2026

GPA: 3.87 · Virginia, USA · Statistical modeling, ML systems, experimental design, supervised/unsupervised learning, data pipelines at scale

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering 2013-2017

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, R, Bash, HTML/CSS

ML & Modeling: scikit-learn, PyTorch, TensorFlow, Random Forest, Logistic Regression, XGBoost, Feature Engineering, Statistical Evaluation, Experimental Design, NumPy, Pandas, BERT, spaCy, Hugging Face

LLM & AI: LangChain, LangGraph, Multi-Provider LLM (Anthropic, OpenAI, Groq, Ollama), RAG Pipelines, ChromaDB, Prompt Engineering, LLM Evaluation, Automated Experimentation, FastAPI

Data & Cloud: AWS (S3, EC2, RDS, Glue/Spark), Azure, PostgreSQL, MySQL, NoSQL, Docker, CI/CD (GitHub Actions), RabbitMQ

Testing & Quality: TDD/BDD, Integration Testing, Git, JIRA, Postman, Agile/Scrum